Toxoplasmosis and mental disorders in the Russian Federation

(Protocol of the investigation)

Objectives

The purpose of the study was to identify the level of Toxoplasma seroprevalence in patients with verified diagnoses of schizophrenia in comparison to healthy people in Moscow City and in the Moscow region.

Background

The association of chronic toxoplasmosis with mental disorders in general and with schizophrenia in particular was noticed in the mid-1950s. In subsequent years, the role of Toxoplasma gondii was established based on its ability to survive for long periods in the nerve cells of the brain. As a consequence, acute manifestations of the infection lead to psychopathic symptoms similar to schizophrenia and hallucinations. In the former USSR, and in other parts of the world, a number of studies were performed with respect to the association of latent toxoplasmosis and schizophrenia. However, with the dissolution of the USSR at the beginning of the 1990s, studies on the subject were halted due to financial problems and have resumed only recently. The reasons for the resumption of such studies in contemporary Russia are related to the progressively increasing incidence of schizophrenia over the last 25-30 years in the country. According to official data, approximately 550 000 persons reported suffering from the disease in 2014. There are reasons to believe that this is only a fraction of the real burden of the disease. Economically, it cost the state no less than approximately US $10 billion. According to the World Health Organization (WHO), there are approximately 24 million persons in the world afflicted by the disease. The highest incidence of the disease was found in Sweden (1.7%), Ireland (1.2%), Russia (0.82%), and the USA (0.72%).

Study population and methods of recruitment

The study was performed at the Sechenov First Moscow State Medical University during 2016. To establish the general prevalence of chronic toxoplasmosis among residents of Moscow city, examinations were carried out on patients attending the Outpatient Department at the Clinical Center of the Sechenov First Moscow State Medical University. No special criteria were selected for examination in terms of age, sex or occupation. Total of 307 persons had been examined.

The main study represented by two groups: experimental and control.

The experimental group consisted of psychiatric patients with a diagnosis of schizophrenia. All members of the experimental group were patients of the Psychiatry Clinic of Sechenov University. The criteria for inclusion were as follows: a) neuropsychiatric disorder diagnosis (schizophrenia); and b) aged 18–45 years. The criterion for exclusion was infectious pathologies (hepatitis, AIDS). A total of 155 patients constituted the experimental group, with 75 men and 80 women (Table 1. The control group consisted of 152 healthy persons aged 18–45 years (82 men and 70 women) who were undergoing routine medical examinations at the Clinical Centre of Sechenov University. The criterion for exclusion was mental disorders in the anamnesis. Participants in the experimental and control groups were informed about the purposes of the study, and informed consent was obtained before enrollment in the study.

Study design

An analytical epidemiological “case-control” study was carried out. All study patients tested for the presence of IgG, avidity antibodies IgG and IgM specific antibodies to *Toxoplasma gondii*. The determination of specific immunoglobulin G and M in the blood serum of the study groups (experimental and control) were determined using “Vector Toxo-IgG” and “Vector Toxo-IgM”, Vector Toxo-IgG-avidity enzyme-linked immunosorbent assay (ELISA) test kits (Vector-Best, Novosibirsk, Russian Federation). The determination of the concentration of IgG to Toxoplasma gondii in the test samples was measured in international units of IU/ml using calibration graphs. A sample was considered negative if the T. gondii IgG level was under 10 IU/ml, and values above 10 IU/ml were considered positive. The avidity was calculated by the following formula: AI (%) = (OD diss./OD dir.)x100 (where AI is the avidity index, OD diss. is the optical density of the dissociating sample, and OD dir. is the direct sample optical density). Evaluation of the results of the enzyme immunoassay for the determination of IgMs was carried out by calculating the coefficient of positivity. The diagnostic significance of chronic toxoplasmosis was the combination of a positive IgG and a negative IgM test, as well as the presence of high avidity antibodies. High avidity antibodies were determined by an avidity index of more than 40%.

Appendix 1 provides a schematic study design.

Statistical analysis

Statistical significance of the results both in the experimental and control groups was obtained through the use of χ^2 criterion, and calculated the odds ratio (OR) with a level of reliability not less than 95%. The Statistical Package EpiInfo was employed for calculations.

To assess the strength of the connection between the presence of toxoplasmosis in the experimental and control groups, and the belonging to the compared groups, we calculated the Pearson correlation, partial correlation, and Mantel-Haenszel common odds ratio estimate. We carried out Multivariate Analysis.

Ethics

The study was approved by the Research Ethics Board of Health of the Sechenov First Moscow State Medical University (protocol № 04-13, 10.04.2013). Participants in the experimental and control groups were informed about the purposes of the study, and informed consent in verbal form was obtained before enrollment in the study.

Results

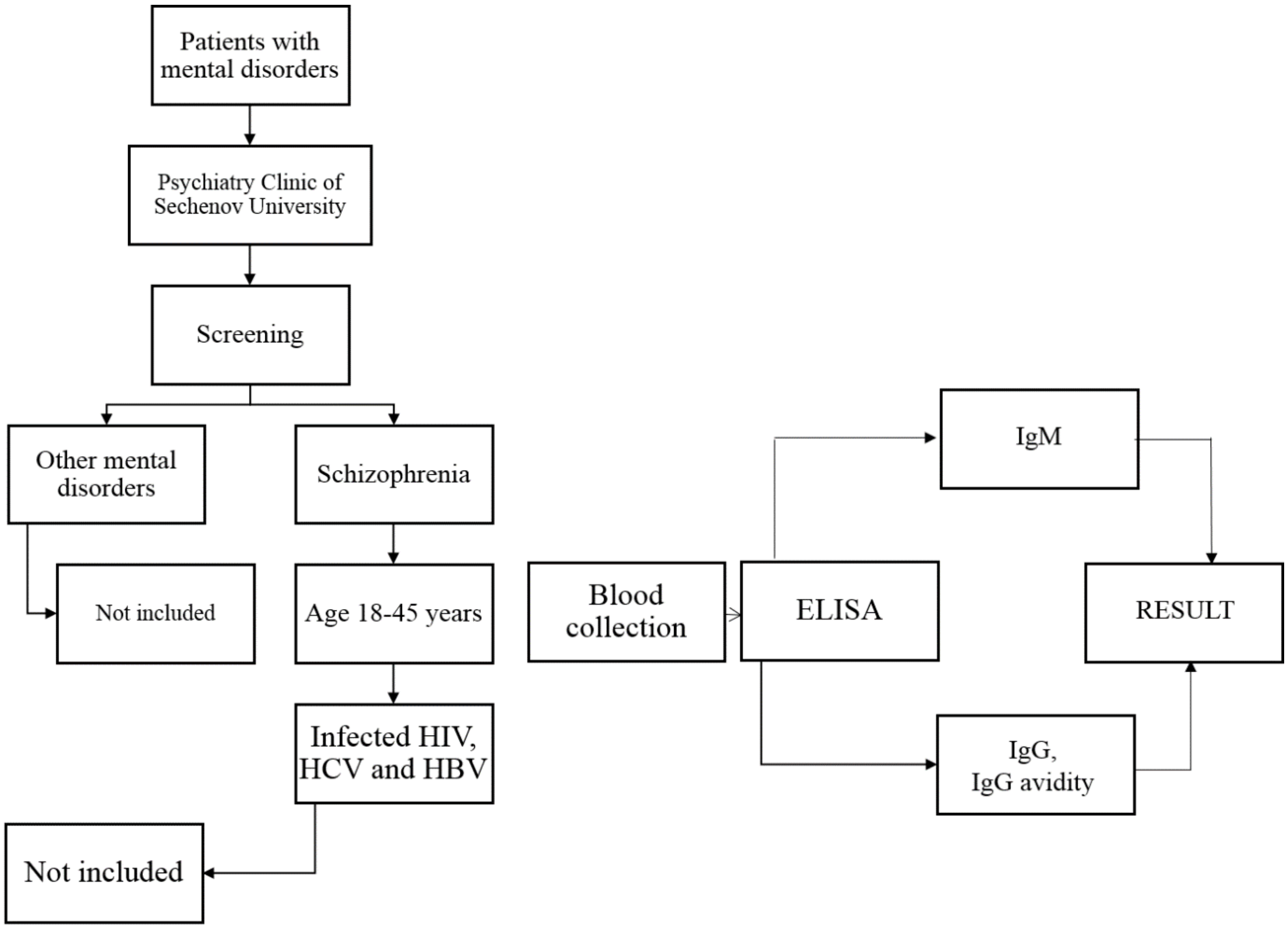
The results of the examination of blood serum in the experimental and control groups are presented in Appendices 2 and 3.

As seen in Table (Appendix 3), the immunoglobulin M was absent in both groups, whereas immunoglobulin G was present in 40% of those tested in the experimental group compared to 25% in the control group. All samples in both groups were tested for Toxoplasma IgG avidity. The avidity is higher than 60% indicates an old infection. The absence of the immunoglobulin IgM in conjunction with the presence of IgG suggests the presence of persons with exclusively chronic form of toxoplasmosis in both the experimental and control groups. It was found that the number of seropositive subjects with IgG in the experimental group was significantly higher than that in the control group. Thus, among the patients with neuropsychiatric disorders, the incidence of cases with chronic toxoplasmosis was more than twice than that in the control group.

Results of Multivariate Analysis show (Appendix 2) that the effect of toxoplasmosis was significant and similar for men (OR=1.72, CI95=0.88-3.37, $p<0.11$) and women (OR=2.19, CI95=1.08-4.43, $p<0.03$). There is statistically significant difference in the ratio of seropositive men ($p<0.11$) and women

($p<0.03$) in the experimental and control groups. Otherwise, there was practically no difference in the prevalence of chronic toxoplasmosis among men and women in the control group (27% and 24%, respectively). Almost the same trend was found in the experimental group among men and women, although the level of prevalence was almost 1.5 times higher compared with the control group.

Appendix 1. The block diagram of the study design.



Appendix 2. Comparative Gender Prevalence of Chronic Toxoplasmosis in the Experimental Group and the Control Group.

Group	Case			Control			Odds ratio	C.I.95	p-values
	Examined	Positive results		Examined	Positive results				
		Absolute number	Percent (%)		Absolute number	Percent (%)			
Men	75	29	39%	82	22	27%	1.72	0.88-3.37	0.11
Women	80	33	41%	70	17	24%	2.19	1.08-4.43	0.03

Appendix 3. Comparative Prevalence of Immunoglobulins M and G to Toxoplasma gondii in the Experimental Group and the Control Group.

Group	Examined	Absolute number		Positive results (%)	Odds ratio	C.I.95	p-values
		IgM	IgG				
Case	155	0	62	40%	1.93	1.16-3.23	0.007
Control	152	0	39	25%			

Appendix 4. The Results of Multivariate Analysis.

Factor	Total (307)	Experimental (155)	Control (152)	p-values
Men	157 51,14%	75 48,39%	82 53,95%	p=0,33
Women	150 48,86%	80 51,610%	70 46,01%	
Age	33,59 SE=0,42 SD=7,41	31,72 SE=0,62 SD=7,66	35,5 SE=0,54 SD=6,66	p=0,000006

The data in table (Appendix 4) show that there was statistically significant difference between the studied groups with respect to sex and age. The compared groups were matched by gender. Despite the fact that the share of men in the "cases" group was less than in the "controls" group (48.39% vs. 53.95%), the use of the χ^2 test showed that the differences were not statistically significant (p =0.33). The average age of the total sample was 33.59 ± 0.42. The average age of the "cases" group was less than the "controls" (31.72 ± 0.62 vs.35.5 ± 0.54). The difference is statistically significant (p = 0.000006, ANOVA test).

Appendix 5. The Results of Gender Analysis.

sex			results		total	p-values	OR 95% CI
			positive	negative			
female	case	N	33	47	80	0,028	2,19 1,08-4,43
		%	41,3%	58,8%	100,0%		
	control	N	17	53	70		
		%	24,3%	75,7%	100,0%		
	total	N	50	100	150		
		%	33,3%	66,7%	100,0%		
male	case	N	29	46	75	0,114	1,72 0,88-3,37
		%	38,7%	61,3%	100,0%		
	control	N	22	60	82		
		%	26,8%	73,2%	100,0%		
	total	N	51	106	157		
		%	32,5%	67,5%	100,0%		

both gender	case	N	62	93	155	0,007	Crude
		%	40,0%	60,0%	100,0%		1,932 1,189-3,189
	control	N	39	113	152		
		%	25,7%	74,3%	100,0%		Adjusted(MH)
	total	N	101	206	307		1,932 1,188-3,142
		%	32,9%	67,1%	100,0%		

A positive relationship was found between the “cases” and the frequency of seropositive toxoplasmosis tests (OR = 1.932, 95% CI 1,189-3,189). The association has been identified in both men and women groups. However, in women it was more statistically significant (OR = 2.19, 95% CI 1.08-4.43, p=0.028). In men, the differences between “cases” and “controls” were less and not statistically significant (OR = 1.72, 1.08-4.43, p = 0.114). To dispose the possible effect of gender on the result, the Mantel-Haenszel test was applied. OR adjusted for sex (1,932 95% CI 1,188-3,142) almost matched the Crude result.

Appendix 6. Multivariate logistic regression analysis

	B (SE)	p-value	OR	95% CI
Constant	-1,402 (0,655)	0,032	0,246	-
case/control	0,694 (0,257)	0,007	2,001	1,210 - 3,310
sex	0,009 (0,247)	0,970	1,009	0,622 - 1,637
age	0,009 (0,017)	0,584	1,009	0,976 - 1,044

Note. R2 = 0,088 (Cox & Snell), 0,118 (Nagelkerke), χ^2 criterion18,987, p=0,015.

Multivariate logistic regression analysis with the result of the laboratory examination as an output variable and the presence of schizophrenia, gender and age as predictors showed that only belonging to the “cases” group is statistically significant with toxoplasma (OR = 2.001, 95% CI: 1.210-3.31, p = 0.007).

Appendix 7. Experimental Group

N of sample	sex	results	age
1	female	negative	26
2	female	negative	25
3	female	negative	34
4	female	positive	32
5	female	positive	40
6	female	positive	37
7	female	positive	37
8	female	positive	37
9	female	positive	26
10	female	negative	20
11	male	negative	45
12	male	negative	33
13	male	positive	40
14	male	negative	31
15	male	negative	23
16	male	negative	28
17	male	negative	25
18	male	negative	37
19	female	positive	35
20	female	negative	33
21	female	negative	29
22	female	negative	44
23	female	negative	45
24	female	positive	42
25	female	negative	38
26	female	positive	29
27	female	positive	23
28	female	positive	29
29	female	negative	19
30	female	negative	23
31	female	negative	39
32	female	negative	38
33	female	positive	27
34	female	negative	35
35	female	positive	20
36	female	positive	24
37	female	negative	40
38	female	positive	18
39	female	positive	33

40	female	positive	37
41	female	positive	27
42	female	negative	32
43	female	negative	35
44	female	positive	28
45	female	positive	37
46	female	negative	31
47	female	positive	40
48	female	negative	36
49	male	negative	36
50	male	negative	23
51	male	negative	42
52	male	positive	26
53	male	negative	39
54	male	positive	21
55	female	negative	45
56	female	negative	38
57	female	negative	20
58	female	negative	34
59	female	positive	41
60	female	negative	30
61	female	negative	26
62	female	negative	23
63	female	positive	20
64	female	negative	35
65	female	positive	41
66	female	positive	23
67	female	negative	39
68	male	positive	31
69	male	positive	33
70	male	negative	45
71	male	negative	38
72	female	positive	34
73	female	negative	33
74	female	negative	34
75	female	negative	44
76	female	negative	36
77	female	negative	42
78	female	positive	31
79	male	positive	38
80	male	positive	20

81	male	positive	34
82	male	positive	41
83	male	positive	39
84	male	negative	27
85	male	negative	42
86	male	negative	19
87	male	negative	39
88	male	positive	38
89	male	positive	45
90	male	negative	29
91	male	negative	29
92	female	negative	38
93	female	positive	29
94	female	negative	29
95	female	negative	29
96	male	negative	38
97	male	negative	20
98	male	negative	34
99	male	negative	41
100	male	negative	30
101	male	positive	35
102	male	negative	36
103	male	negative	39
104	male	negative	27
105	male	negative	42
106	female	negative	26
107	female	negative	23
108	female	negative	35
109	female	negative	20
110	female	positive	26
111	female	negative	25
112	female	negative	27
113	female	positive	26
114	male	negative	19
115	male	negative	39
116	male	positive	38
117	male	negative	45
118	male	negative	29
119	male	negative	29
120	male	negative	38
121	male	negative	34

122	male	positive	20
123	male	negative	20
124	male	positive	24
125	male	negative	31
126	male	negative	16
127	female	negative	19
128	female	negative	22
129	female	positive	43
130	female	negative	23
131	female	negative	37
132	female	positive	38
133	female	positive	26
134	female	negative	23
135	male	negative	20
136	male	negative	35
137	male	positive	41
138	male	positive	23
139	male	positive	37
140	male	positive	25
141	male	positive	42
142	male	negative	28
143	male	negative	24
144	male	positive	23
145	male	positive	28
146	male	positive	41
147	male	positive	42
148	male	positive	28
149	male	positive	36
150	male	positive	29
151	male	negative	26
152	male	positive	31
153	male	negative	42
154	male	negative	19
155	male	negative	23

Appendix 8. Control Group

N of sample	sex	results	age
1	male	positive	40
2	male	positive	41
3	male	negative	34
4	male	negative	38
5	male	negative	34
6	male	negative	26
7	male	negative	30
8	male	negative	34
9	male	negative	41
10	male	negative	34
11	male	negative	34
12	male	negative	38
13	male	positive	34
14	male	positive	41
15	male	positive	45
16	male	positive	28
17	male	negative	45
18	male	negative	45
19	male	negative	32
20	male	negative	45
21	male	negative	45
22	male	negative	45
23	male	negative	42
24	male	negative	36
25	male	negative	41
26	male	negative	42
27	male	negative	42
28	male	negative	42
29	male	negative	38
30	male	negative	39
31	male	negative	34
32	male	positive	39
33	male	negative	32
34	male	negative	39
35	male	negative	39
36	male	negative	39
37	male	positive	27
38	male	positive	31
39	male	positive	43

40	male	positive	41
41	male	positive	41
42	male	negative	43
43	male	negative	43
44	male	negative	41
45	male	negative	43
46	male	negative	20
47	male	negative	41
48	male	negative	37
49	male	negative	43
50	male	negative	32
51	male	negative	29
52	male	negative	31
53	male	negative	26
54	male	negative	43
55	male	negative	43
56	male	negative	26
57	male	negative	43
58	male	positive	43
59	male	positive	29
60	male	positive	43
61	male	positive	43
62	male	positive	30
63	male	negative	30
64	male	negative	30
65	male	positive	26
66	male	positive	26
67	male	positive	30
68	male	negative	30
69	male	negative	28
70	male	negative	37
71	male	negative	31
72	male	negative	30
73	male	negative	41
74	male	negative	30
75	male	negative	27
76	male	positive	30
77	male	positive	30
78	male	negative	21
79	male	negative	39
80	male	negative	30

81	male	negative	41
82	male	negative	30
83	female	positive	30
84	female	negative	30
85	female	positive	28
86	female	negative	37
87	female	negative	31
88	female	negative	30
89	female	negative	41
90	female	negative	30
91	female	negative	27
92	female	negative	30
93	female	positive	30
94	female	negative	21
95	female	negative	39
96	female	negative	30
97	female	negative	41
98	female	positive	30
99	female	positive	41
100	female	negative	43
101	female	negative	20
102	female	negative	41
103	female	negative	37
104	female	negative	43
105	female	negative	32
106	female	positive	29
107	female	negative	31
108	female	negative	26
109	female	negative	43
110	female	positive	43
111	female	negative	26
112	female	positive	43
113	female	negative	43
114	female	negative	29
115	female	negative	43
116	female	positive	43
117	female	negative	30
118	female	negative	42
119	female	positive	36
120	female	negative	41
121	female	negative	42

122	female	negative	42
123	female	positive	42
124	female	negative	38
125	female	negative	39
126	female	positive	34
127	female	negative	39
128	female	negative	32
129	female	negative	41
130	female	negative	43
131	female	negative	20
132	female	positive	41
133	female	negative	37
134	female	negative	43
135	female	positive	32
136	female	negative	29
137	female	negative	31
138	female	negative	26
139	female	negative	43
140	female	negative	43
141	female	negative	26
142	female	positive	43
143	female	negative	43
144	female	negative	29
145	female	negative	36
146	female	positive	31
147	female	negative	29
148	female	negative	37
149	female	negative	34
150	female	negative	22
151	female	positive	41
152	female	negative	33